

● HARD

● GEOMETRY AND TRIGONOMETRY

Circles

10 questions · ~15 min

03

SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

A. $x - 8^2 + y - 8^2 = 16$

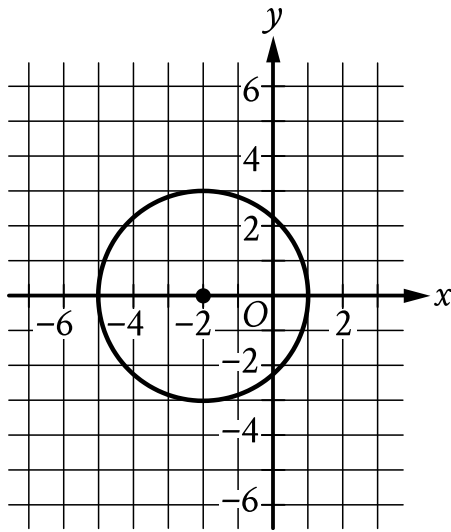
B. $x - 8^2 + y - 4^2 = 16$

C. $x - 4^2 + y - 9^2 = 16$

D. $x^2 + y - 9^2 = 16$

In the xy -plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

- A. 5
- B. 6.5
- C. $\sqrt{40}$
- D. $\sqrt{50}$



- The center of the circle is the point (negative 2 comma 0).
- Clockwise from top, the circle passes through the following points:
 - (negative 2 comma 3)
 - (1 comma 0)
 - (negative 2 comma negative 3)
 - (negative 5 comma 0)

Circle A (shown) is defined by the equation $x + 2^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A . Which equation defines circle B ?

- A. $x + 2^2 + y + 6^2 = 49$
- B. $2x + 2^2 + 2y + 6^2 = 9$
- C. $x + 2^2 + y - 6^2 = 49$
- D. $2x + 2^2 + 2y - 6^2 = 9$

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

The equation $(x+6)^2 + (y+3)^2 = 121$ defines a circle in the xy-plane. What is the radius of the circle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A circle in the xy -plane has equation $(x+3)^2 + (y-1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

- A. $(-7, 3)$
- B. $(-3, 1)$
- C. $(0, 0)$
- D. $(3, 2)$

What is the diameter of the circle in the xy -plane with equation $x - 5^2 + y - 3^2 = 16$?

- A. 4
- B. 8
- C. 16
- D. 32

Q7

GRID-IN GEOMETRY AND TRIGONOMETRY

Circle A in the xy -plane has the equation $x + 5^2 + y - 5^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $x + 5^2 + y - 5^2 = k$, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

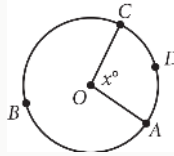
A circle in the xy -plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

A. $(x - 19)^2 + (y - 18)^2 = 81k$

B. $(x - 19)^2 + (y - 18)^2 = 81k^2$

C. $(x - 19)^2 + (y - 18)^2 = 9k$

D. $(x - 19)^2 + (y - 18)^2 = 9k^2$



The circle above has center O, the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

- A. 3
- B. 6
- C. 9
- D. 24